Sample Technical Document for RAG Testing

Purpose

This sample PDF is designed for testing a document-upload pipeline. It contains searchable text that can later be extracted, divided into chunks, converted into embeddings, and retrieved to answer questions.

1. Document Management

A document ingestion system should record the original filename, stored filename, content type, file size, upload timestamp, and processing status. The original filename is useful for display, while a generated unique filename helps avoid collisions in storage.

2. PDF Validation

A basic PDF upload check can verify that the file is not empty, that its size is within the configured limit, that its filename ends in .pdf, and that its first five bytes contain the PDF signature percent-PDF-hyphen. These checks are useful but do not prove that the entire file is valid or safe.

3. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, retrieves relevant information from a knowledge source and provides it as context to a language model. A typical pipeline loads a document, extracts text, cleans the text, splits it into chunks, generates embeddings, stores vectors, retrieves relevant chunks for a query, and produces an answer grounded in the retrieved content.

4. Chunking and Embeddings

Chunking divides long documents into smaller text sections. Chunk size and overlap affect retrieval quality. Embeddings represent text as numerical vectors so semantically related content can be found using vector similarity. A vector database can store these vectors alongside metadata such as document ID, page number, and chunk index.

5. Questions to Test Retrieval

Question A: What metadata should a document ingestion system record? Question B: What does RAG stand for? Question C: Why are documents split into chunks? Question D: What is the purpose of embeddings?

End of sample document. This file contains instructional content and is intended for software development and testing.